

L5: Clustering in Weighted Networks

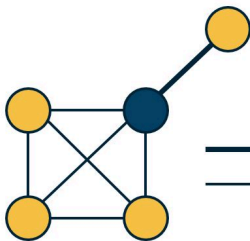
The definition of clustering coefficient can also be generalized for weighted networks as follows. Suppose that $w_{i,j}$ is the weight of the edge between nodes i and j .

First, in weighted networks, instead of the degree of a node we often talk about its “strength”, defined as the sum of all weights of the node’s connections:

$$s_i = \sum_j A_{i,j} w_{i,j}$$

Then, the weighted clustering coefficient of node i is defined as:

$$C_w(i) = \frac{1}{s_i (k_i - 1)} \sum_{j,h} \frac{w_{i,j} + w_{i,h}}{2} A_{i,j} A_{i,h} A_{j,h}$$



$$C = 0.5 \quad C_w = 0.25$$

The normalization term $\frac{1}{s_i (k_i - 1)}$ is such that the maximum value of the weighted clustering coefficient is one.

The product of the three adjacency matrix elements at the right is one only if the nodes i, j, h form a triangle. In that case, that triangle contributes to the clustering coefficient of node- i based on the average weight of the two edges that connect node i with j and h , respectively. Note that the weight between nodes j and h does not matter.

The visualization shows the unweighted and weighted clustering coefficient values for the darker node. That node has a stronger connection with a node that does not belong to the cluster of nodes at the lower-left side.

This is why the weighted clustering coefficient is lower than the unweighted.

Food For Thought

How would you generalize the definition of Transitivity for weighted networks?